

Yoonseok Choi

Seoul, South Korea

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Biography

I am a Ph.D. student at Yonsei University, advised by **Prof. Dong-Hyun Kim** in the **Medical Imaging AI Lab (MI-LAB)**. In 2025, I worked as a **visiting student** at the **Vision and Learning Lab (VLLab), UC Merced**, under the supervision of **Prof. Ming-Hsuan Yang**, broadening my research into generative modeling and image reconstruction.

My research addresses the **missing modality problem** in MRI, where incomplete acquisitions are common in clinical practice yet severely limit downstream analysis. I develop methods that leverage knowledge distillation, generative modeling, and disentangled representation learning to enable reliable diagnosis from incomplete data. These techniques, learning from incomplete data and recovering missing information, apply broadly across computer vision. My wider interests include super-resolution, motion artifact correction, and segmentation in brain MRI.

Education

Ph.D. in Electrical and Electronic Engineering

Yonsei University

Advisor: Dong-Hyun Kim

Seoul, South Korea

Mar 2023 – Present

M.S. in Electrical and Electronic Engineering

Yonsei University

Advisor: Dong-Hyun Kim

Seoul, South Korea

Mar 2021 – Feb 2023

B.S. in Biomedical Engineering

Yonsei University

Advisor: Young-Ro Yoon

Wonju, South Korea

Mar 2015 – Feb 2021

Selected Publications (* Equal Contribution)

Conference Paper

1. DRIFT: Difficulty-aware Rectified Flows for Through-plane MRI Super-Resolution

Yoonseok Choi, Eun-Gyu Ha, Daniel Kim, Mohammed A. Al-masni, Ming-Hsuan Yang, and Dong-Hyun Kim
ECCV 2026

2. Controlling Motion Transfer in Diffusion Transformers via Attention Heads

Sunyoung Jung*, Jiwoo Park*, **Yoonseok Choi**, Kyobin Choo, Ming-Hsuan Yang, and Seong Jae Hwang
ECCV 2026

3. TESLA: Test-time Reference-free Through-plane Super-resolution for Multi-contrast Brain MRI

Yoonseok Choi, Sunyoung Jung, Mohammed A. Al-masni, Ming-Hsuan Yang, and Dong-Hyun Kim
MICCAI 2025, Oral presentation, Top 2.2%

4. Deformation-Aware Segmentation Network Robust to Motion Artifacts for Brain Tissue Segmentation using Disentanglement Learning

Sunyoung Jung, **Yoonseok Choi**, Mohammed A. Al-masni, Minyoung Jung, and Dong-Hyun Kim
MICCAI 2024

5. Brain Tissue Segmentation Robust to motion artifacts using Deformation-Aware Network

Sunyoung Jung, **Yoonseok Choi**, Mohammed A. Al-masni, and Dong-Hyun Kim
ISMRM 2024, Oral presentation

6. Two-Stage Deep Learning with Multi-Pathway Network for Brain Tumor Segmentation and Malignancy Identification From MR Images

Yoonseok Choi, Mohammed A. Al-masni, Hyeok Park, Jun-ho Kim, and Dong-Hyun Kim

ISMRM 2023, Oral presentation

7. 3D CMM-Net with Deeper Encoder for Semantic Segmentation of Brain Tumors in BraTS2021 Challenge

Yoonseok Choi, Mohammed A. Al-masni, and Dong-Hyun Kim

MICCAI 2021 Brain Lesion Workshop

Journal Paper

1. A Single Stage Knowledge Distillation Network for Brain Tumor Segmentation on Limited MR Image Modalities

Yoonseok Choi, Mohammed A. Al-masni, Kyu-Jin Jung, Roh-Eul Yoo, Seong-Yeong Lee, and Dong-Hyun Kim

Computer Methods and Programs in Biomedicine (CMPB), 2023, impact factor 6.1

Patents

1. Apparatus and Method for Segmenting Brain Tumors from MR Images

Yoonseok Choi and Dong-Hyun Kim

Registration Number: 10-2867230-0000, Registration Date: Sep 26, 2025

2. Integrated Software Platform to Visualize Brain Tumor Segmentation Masks from MR Image

Yoonseok Choi, Hyeok Park, and Dong-Hyun Kim

Registration Number: C-2022-032255 (Software Registration), Registration Date: Aug 17, 2022

Experience

Vision and Learning Lab (VLLab) - Visiting Student

University of California at Merced (Advisor: Ming-Hsuan Yang)

Merced, USA

Feb 2025 – Jan 2026

- Research on super-resolution and diffusion models in medical imaging

Medical Imaging Artificial Intelligence Lab (MILab) - Intern

Yonsei University (Advisor: Dong-Hyun Kim)

Seoul, South Korea

Jul 2020 – Feb 2021

- Research on parallel imaging and semantic segmentation in Brain MRI

Biomedical Signal Processing Lab - Intern

Yonsei University (Advisor: Young-Ro Yoon)

Wonju, South Korea

Mar 2019 – Dec 2019

- Development of a skin condition measurement device using TEWL (TransEpidermal Water Loss)

Awards and Honors

1. Cognitive and Biological Factors Related to the Development of Question-Asking Abilities in School-Aged Children, Junior Convergence Research Group (1st place, 1,500,000 won), 2025

Hyebin Sung, Seoran Kim, Yuju Shin, Dongwook Kim, Jae-Yoon Kim, Jun-Ho Kim, Soohyoung Lee, **Yoonseok Choi**, and Eun-Gyu Ha

2. BK21 FOUR (Brain Korea 21 Four) Project; Support Program for Outstanding Graduate Students' International Joint Training 1 year from the commencement of training (12months, 26,000,000 won), 2024

Yoonseok Choi

3. Structural brain correlates of foreign language proficiency and experiences, Junior Convergence Research Group (3rd place, 500,000 won), 2023

XIAOQIAO WANG, Seoran Kim, Jae-Yoon Kim, Jun-Ho Kim, **Yoonseok Choi**, and Eun-Gyu Ha

4. DSU-Net2D: Deep Supervision U-Net2D, Medical Image Processing Contest with Rayence and Yonsei University (2nd place, 2,000,000 won), 2022

Yoonseok Choi and Sewook Kim

Academic Activities

Conference Reviewer

<i>The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i>	2026
<i>MEDICAL IMAGE COMPUTING AND COMPUTER ASSISTED INTERVENTION (MICCAI)</i>	2025
<i>International Conference on Computer Vision (ICCV)</i>	2025